

Student Management Dashboard

Comprehensive Business Intelligence & Academic Analytics Documentation

Tool Used: Microsoft Power BI Desktop
Project Type: Academic Analytics Dashboard

Target Audience: Academic Management & Evaluation Board
Status: Final Deployed Version

1. Executive Summary & Purpose

The primary objective of this Student Management Dashboard is to provide a central, interactive, and data-driven platform for educational administrators and faculty members. Instead of relying on static Excel sheets, this dashboard converts raw student datasets into real-time business intelligence insights.

- **Academic Monitoring:** Provides high-level aggregated metrics for student performance, average GPA, and attendance status.
- **Departmental Benchmarking:** Compares scoring patterns across various academic departments to highlight strengths and areas for intervention.
- **Risk Identification:** Pinpoints grade distributions and attendance anomalies to support proactive student support.

2. Data Modeling & Core Architecture

The dashboard connects four relational database tables: Students, Courses, Enrollments_Marks, and Attendance.

- **Relational Keys:** One-to-Many relationships configured using StudentID and CourseID.
- **Dynamic Filtering:** Selecting any slicers (Department, Course, Semester, Attendance) automatically updates all connected KPI cards, column charts, donut charts, and detail tables synchronously.

3. Dynamic DAX Measures

Custom Data Analysis Expressions (DAX) were engineered to replace implicit aggregations, ensuring dynamic calculation performance:

Total Student Count

```
Total Students = COUNT(Students[StudentID])
```

Overall Average Marks

```
Overall Avg Marks = AVERAGE(Enrollments_Marks[Marks])
```

Overall Average GPA

```
Overall Avg GPA = AVERAGE(Students[GPA])
```

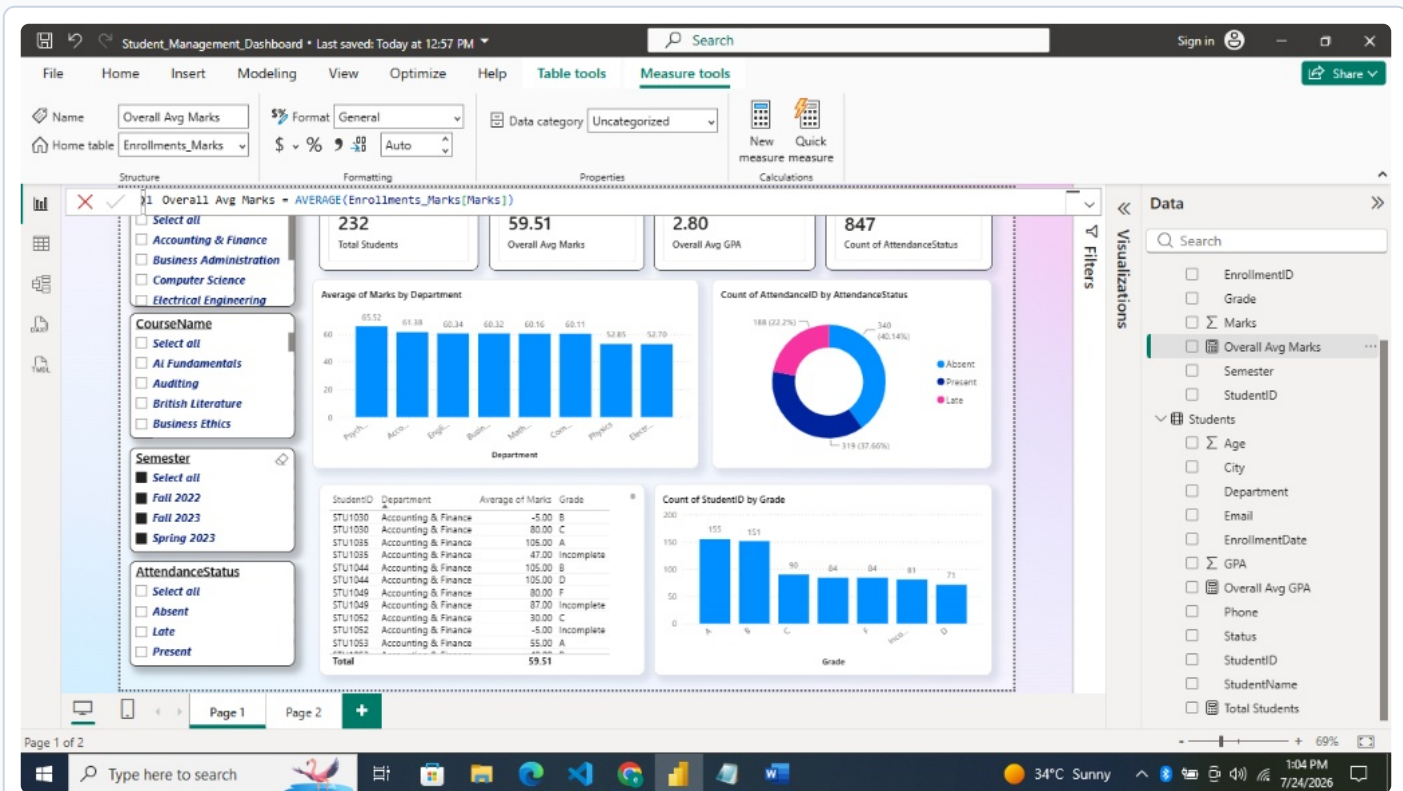


Figure 1: Implementation of Dynamic DAX Measures in Data Pane & Formula Bar

4. Visualizations & Analytical Insights

- KPI Summary Cards (Top Row):** Displays *Total Students*, *Overall Average Marks*, *Overall Average GPA*, and *Total Attendance Entries*.
- Average Marks by Department (Bar Chart):** Analyzes performance across departments (Psychology, Accounting, Computer Science, etc.).
- Attendance Status Breakdown (Donut Chart):** Evaluates percentage split between Present, Absent, and Late entries.
- Grade Distribution (Column Chart):** Displays count of students per grade tier (A, B, C, D, F, Incomplete).
- Student Details Table (Bottom Left):** Granular student-level records for auditability.

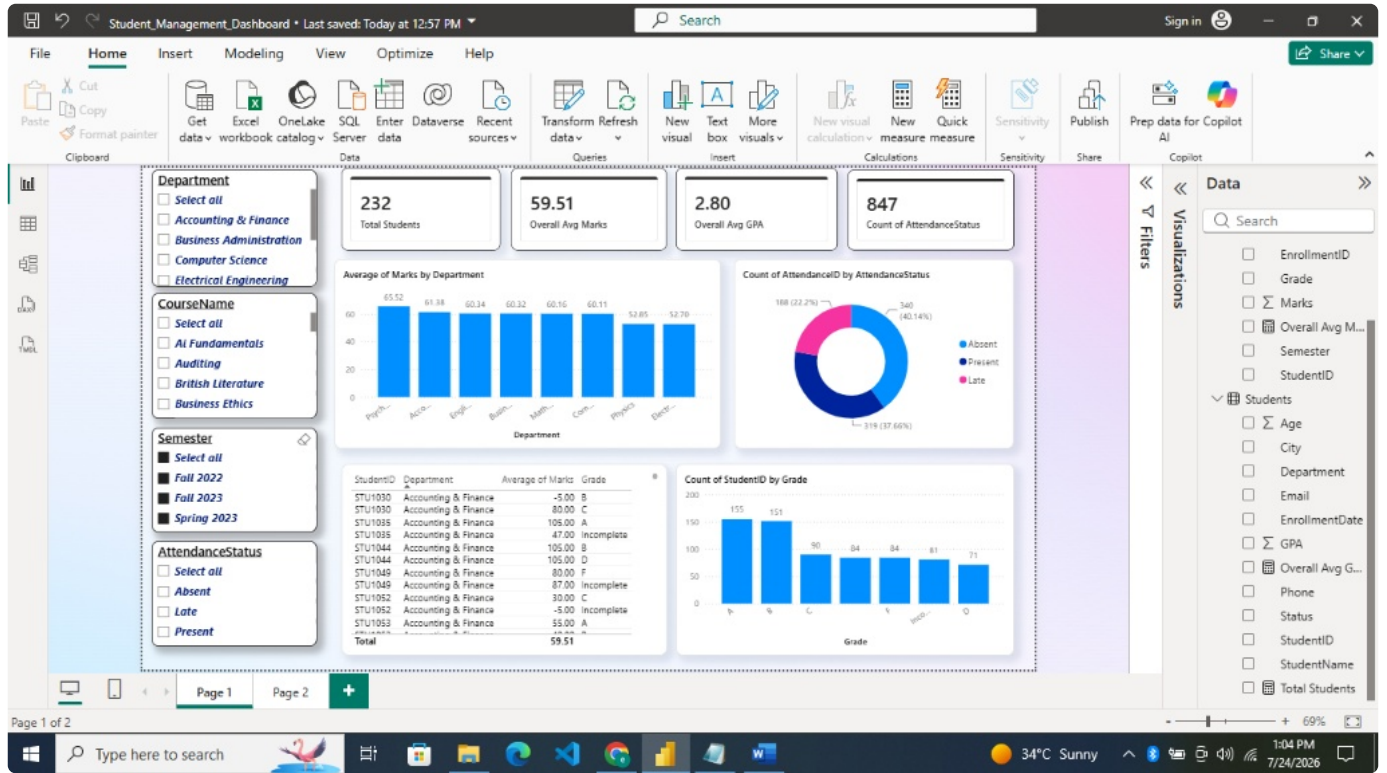


Figure 2: Complete Student Management Dashboard Interface (Page 1)

5. Custom Tooltip Implementation

Concept: A dedicated Tooltip Page (Page 2) was configured and linked to the *Average Marks by Department* bar chart.

Functionality: Hovering over any department bar dynamically renders a pop-up chart showing course-level average marks for that specific department, maintaining canvas efficiency without cluttering Page 1.

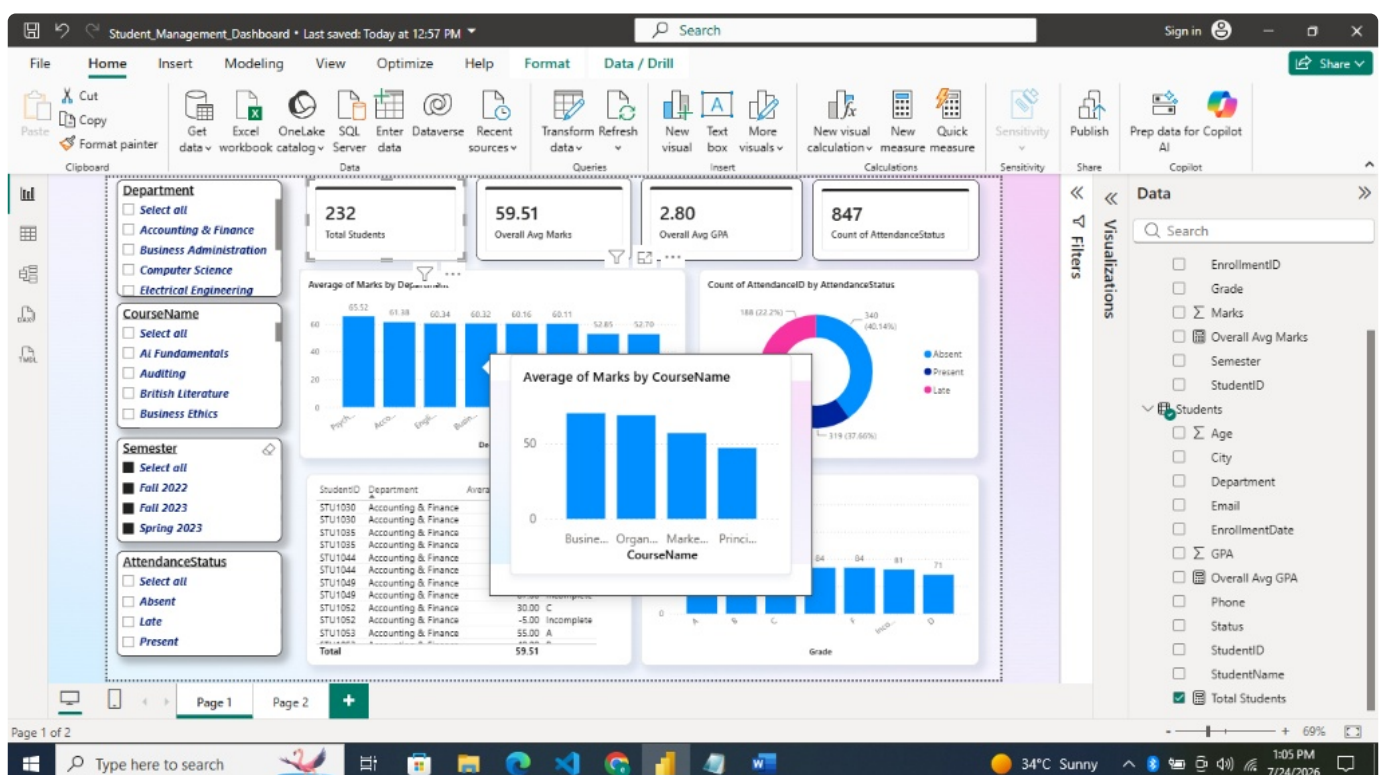


Figure 3: Interactive Department-to-Course Custom Page Tooltip Pop-up

6. UI/UX Design Standards

- **Layout:** Balanced 4-quadrant structured layout with a left-hand filter panel.
- **Styling:** Soft gray background contrast with white container cards, 10px rounded corners, and drop shadows.
- **Typography:** Enhanced font sizes for slicers and headers to meet executive presentation standards.